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Academic Profile

Lecturer-researcher in the Department of Automatic Control, Faculty of Electrical Engineering. Born on 15 August 1989 in Oran; Algerian citizen. My research addresses high-voltage direct-current (HVDC) links, modular multilevel converters, FACTS devices and the integration of renewable energy. Recent work covers artificial intelligence, machine learning, robust and intelligent control, optimisation, microgrids, energy management of smart buildings and nonlinear electromechanical systems. My academic activity combines teaching, research supervision, production of teaching materials and programme leadership.

Positions and Responsibilities

- **Since June 2024: Head of the Automatic Control programme**, Faculty of Electrical Engineering, Djillali Liabès University of Sidi Bel Abbès.
- **Since November 2022: Associate Professor (Class A)**, Department of Automatic Control, Djillali Liabès University of Sidi Bel Abbès.
- **December 2017–October 2022: Lecturer-researcher**, Department of Automatic Control. Recruited as Assistant Professor (Class B) on 31 December 2017, then Associate Professor (Class B).
- **2012–2017: Part-time lecturer**, Department of Electrotechnics, same university: lectures, tutorials and laboratory sessions in power systems, electrical engineering, numerical methods, computer science and signal theory.

Journal Articles and Scientific Publications

DOI links point to the publisher's record. Publications resulting from collaboration with foreign teams or French research partners carry the corresponding note.

1. **2026.** M. A. Bouchelarm, **M. A. Djehaf**, A. Boulenouar, M. Chafi, N. Benseddiq. Regularized radial basis function for mixed-mode fracture parameters prediction using artificial neural networks. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science.* DOI: 10.1177/09544062261478692. *Collaboration: University of Lille (Unité de Mécanique de Lille), France.* [Link]

2. **2026.** **M. A. Djehaf**, M.-H. Benzaama, A. Boudenne, K. Benzarti. A deep residual graph convolutional neural network for hybrid physics data-driven modelling of latent heat transfer in cement mortar walls filled with bio-based phase change materials. *Applied Thermal Engineering.* DOI: 10.1016/j.applthermaleng.2026.132512. *Collaboration: IRC-ESTP, Cachan, France.* [Link]

3. **2026.** A. Feddaoui, **M. A. Djehaf**. A novel rule based control method for hierarchical wind farm power management. *Energy Reports*. DOI: 10.1016/j.egyr.2026.109599. [Link]

4. **2026.** A. Feddaoui, L. Farah, A. Benretem, **M. A. Djehaf**, Y. Djeriri. Enhanced control of DFIG-based wind turbine using hybrid fuzzy super-twisting sliding mode control for improved power quality and robustness. *Journal of Engineering and Technology for Industrial Applications*. DOI: 10.5935/jetia.v12i57.2920. [Link]

5. **2026. M. A. Djehaf**, A. Hamet Sidi, Y. I. Djilani Kobibi. Visual Servoing Sliding Mode Control with Vibration Model Compensation for Trajectory Tracking in a 2-DOF Ball Balancer System. *Vibration*, 9(1), 19. DOI: 10.3390/vibration9010019. [Link]

6. **2026.** M. A. Bouchelarm, **M. A. Djehaf**, A. Boulenouar, M. Chafi, N. Benseddiq. Improving the performance of cracked aluminium structures using composite patch repairs: a numerical and statistical investigation. *The International Journal of Advanced Manufacturing Technology*. DOI: 10.1007/s00170-025-17194-z. *Collaboration: University of Lille (Unité de Mécanique de Lille), France.* [Link]

7. **2026.** K. Djehaf, **M. A. Djehaf**. Temperature seasonality and spatial confounding of cutaneous leishmaniasis in Algeria: a source-verified national panel, 2000–2024. Dataset and reproducibility code, *Zenodo*. DOI: 10.5281/zenodo.21813194. [Link]

8. **2025.** A. Feddaoui, L. Farah, A. Benretem, **M. A. Djehaf**. Grey wolf optimization of PI controller for power management in wind farms: a novel approach. *Turkish Journal of Electrical Engineering and Computer Sciences*. DOI: 10.55730/1300-0632.4156. [Link]

9. **2024.** A. Hamet Sidi, **M. A. Djehaf**, Y. I. Djilani Kobibi. Enhanced Ball Trajectory Tracking Using Visual Servoing with 2-DOF Ball on Plate Balancing System. *Journal Européen des Systèmes Automatisés*, 57(2). DOI: 10.18280/jesa.570210. [Link]

10. **2024.** A. Achar, Y. Djeriri, H. Benbouhenni, A. Bentaallah, **M. A. Djehaf**, S. Hanafi, I. Colak. Robust Lyapunov-Based Power Controllers with Integral Action for a Wind Farm. *Electric Power Components and Systems*. DOI: 10.1080/15325008.2024.2311880. *Collaboration: Nisantasi University, Istanbul, Turkey.* [Link]

11. **2024.** H. Boudra, R. Sayah, K. Djehaf, **M. A. Djehaf**, A. Benaissa. Investigating Multi-Needle Electrode Configurations for Improved Corona Discharge Charging of Nonwoven Filter Media. *Journal of Electrical Systems*, 20(9s). [Link]

12. **2024.** R. Sayah, H. Boudra, K. Djehaf, **M. A. Djehaf**. Experimental and Statistical Optimization of Electrostatic Separation Using Three-Level Full Factorial Design. *Journal of Electrical Systems*, 20(9s). [Link]

13. **2023.** A. Feddaoui, L. Farah, A. Benretem, **M. A. Djehaf**, Z. Bouguerra, Y. Djeriri. Enhancing power control in doubly fed induction generator based wind turbines: a performance comparison of fuzzy logic and sliding mode control methods. *The Scientific Bulletin of Electrical Engineering Faculty*. DOI: 10.2478/sbeef-2023-0018. [Link]

14. **2023. M. A. Djehaf**, Y. I. Djilani Kobibi, M. Khatir. Dynamic Performance of High-Voltage Direct Current Systems for Offshore Wind Farm Based on Modular Multilevel Converter. *WSEAS Transactions on Power Systems*, 18. DOI: 10.37394/232016.2023.18.12. [Link]

15. **2023.** A. Achar, Y. Djeriri, A. Bentaallah, S. Hanafi, **M. A. Djehaf**, R. Bouddou. Lyapunov-Based Robust Power Controllers for a Wind Farm Using Parallel Multicell Converters. *Przegląd Elektrotechniczny*, 99(4), 247–254. DOI: 10.15199/48.2023.04.43. [Link]

16. **2022.** Y. I. Djilani Kobibi, **M. A. Djehaf**, M. Khatir, M. Ouadafraksou. Continuation Power Flow Analysis of Power System Voltage Stability with Unified Power Flow Controller. *Journal of Intelligent Systems and Control*, 1(1). DOI: 10.56578/jisc010106. [Link]

17. **2022.** Y. I. Djilani Kobibi, **M. A. Djehaf**, M. Khatir. MMC-Based HVDC Transmission Systems Models in EMT-Type Program. *Journal Européen des Systèmes Automatisés*, 55(6). DOI: 10.18280/jesa.550606. [Link]

18. **2022. M. A. Djehaf**, Y. I. Djilani Kobibi, M. Khatir, M. Ouadafraksou. Transient Stability Improvement of Multi Machine Power System Including DFIG Wind Farm Using HVDC Link. *Journal Européen des Systèmes Automatisés*, 55(4). DOI: 10.18280/jesa.550407. [Link]

19. **2021.** M. Belgacem, M. Khatir, **M. A. Djehaf**, S. A. Zidi, R. Bouddou. Implementation of DC voltage controllers on enhancing the stability of multi-terminal DC grids. *International Journal of Electrical and Computer Engineering*, 11(3). DOI: 10.11591/ijece.v11i3.pp1894-1904. [Link]

20. **2016.** Modeling of a Multi-level Converter Based MTDC System for the Study of Power System Steady-State and Transient Characteristics. *Journal of Electrical Engineering*, 16(1), 116–124. [Link]

21. **2015.** Y. I. Djilani Kobibi, S. Hadjeri, **M. A. Djehaf**. Effect of the UPFC on a multimachine power system steady state and dynamic performance. *Journal of Electrical Engineering*, 15(4), 283–288. [Link]

22. **2015. M. A. Djehaf**, S. A. Zidi, M. Khatir, S. Hadjeri, Y. Djilani Kobibi. Research document 7-R62.131 (ResearchGate deposit). DOI: 10.13140/RG.2.1.2924.4240. [Link]

23. **2014. M. A. Djehaf**, S. A. Zidi, Y. Djilani Kobibi. AC Versus DC Link Comparison Based on Power Flow Analysis of a Multimachine Power System. *Leonardo Journal of Sciences*, 24, 61–74. [Link]

24. **2014.** Y. I. Djilani Kobibi, S. Hadjeri, **M. A. Djehaf**. Study of UPFC Optimal Location Considering Loss Reduction and Improvement of Voltage Stability and Power Flow. *Leonardo Journal of Sciences*, 24, 85–100. [Link]

25. **2014. M. A. Djehaf**, S. A. Zidi, Y. Djilani Kobibi. Dynamic Performance of HVDC Link based 3-level VSC Supplying a Passive Load. *International Journal of Advanced Engineering and Science*, 3(1). [Link]

26. **2014.** Y. I. Djilani Kobibi, S. Hadjeri, **M. A. Djehaf**. Modelling a UPFC for the Study of Power System Steady State and Transient Characteristics. *International Journal of Advanced Engineering and Science*, 3(1). [Link]

27. **2015/2016. M. A. Djehaf**, S. A. Zidi, S. Hadjeri, Y. Djilani Kobibi. Steady-state and transient performance of HVDC link based 3-level VSC supplying a passive load. *Synthèse*, 31, 80–89. [Link]

Conference Papers (Indexed Proceedings Available Online)

1. **2026.** C. Larouci, M. W. Omari, N. Rizoug, **M. A. Djehaf**, C. Morel. Étude comparative des techniques de refroidissement pour les modules de puissance IGBT pour véhicules électriques [A comparative study of cooling techniques for IGBT power modules in electric vehicles; in French]. *Jeunes Chercheurs en Génie Électrique (JCGE 2026). Collaboration: ESTACA'Lab, France.* [Link]

2. **2026.** M. W. Omari, N. Rizoug, **M. A. Djehaf**, C. Morel, C. Larouci. A Comparative Analysis of Cooling Techniques for IGBT power modules in electric vehicles. *CoDIT 2026*. DOI: 10.1109/CODIT70676.2026.11630975. *Collaboration: ESTACA'Lab, France.* [Link]

3. **2026.** W. Khiat, **M. A. Djehaf**. Modeling, Control, and Performance Evaluation of MMC-Based STATCOM Using SSBC Topology. *ElCon-PEIA 2026*. DOI: 10.1109/ELCON-PEIA69858.2026.11451865. [Link]

4. **2025.** Optimized Climate-Resilient Desalination for Algeria's 2030 Water Strategy. *6th International Electronic Conference on Applied Sciences.* [Link]

5. **2025.** Real-Time Gain Scheduling via Adaptive Fuzzy PID Control: Application to Nonlinear Inverted Pendulum Stabilization. *6th International Electronic Conference on Applied Sciences.* [Link]

6. **2024.** Enhancing the Algerian Power System for Long-Distance Transmission: A Comprehensive Study on the Implementation of HVDC Technology. *5th International Electronic Conference on Applied Sciences.* [Link]

7. **2024.** A. Amadou Adamou, **M. A. Djehaf**, Y. I. Djilani Kobibi, M. Rezkallah, A. Feddaoui. Optimizing Grid-Connected Microgrid Energy Management Through Advanced Heuristic Algorithms. *ICAEE.* DOI: 10.1109/ICAEE61760.2024.10783352. *Collaboration: Research Center in Energy Intelligence and Innovation (CR2Ie), Québec, Canada.* [Link]

8. **2024.** O. Fezazi, A. Semmak, H. Köten, **M. A. Djehaf**, A. N. E. Ayad, B. B. Mallik. RTLAB-Based Performance Validation of PV-Powered Single-Phase Asynchronous Motor Pumps. *ICAEE.* DOI: 10.1109/ICAEE61760.2024.10783175. *Collaboration: Istanbul Medeniyet University, Turkey; University of Engineering & Management, Kolkata, India.* [Link]

9. **2024.** A. Feddaoui, **M. A. Djehaf**, B. Bouchelarm, Y. I. Djilani Kobibi, Y. Djeriri. Sliding Mode Control for Underactuated Nonlinear Mechanical Systems. *ICAEE.* DOI: 10.1109/ICAEE61760.2024.10783171. [Link]

10. **2024. M. A. Djehaf**, A. Hamet Sidi, Y. I. Djilani Kobibi, O. Fezazi, Y. Djeriri. Visual Servoing-Enhanced Optimal Control for Trajectory Tracking in 2-DOF Ball on Plate Systems. *ICAEE.* DOI: 10.1109/ICAEE61760.2024.10783220. [Link]

11. **2020.** M. Belgacem, M. Khatir, **M. A. Djehaf**, R. Bouddou, F. Behamida. Analysis and impacts of implementing droop control in Multi-Terminal VSC-HVDC under steady state and transient stability. *ICEIT.* DOI: 10.1109/ICEIT48248.2020.9113200. [Link]

12. **2019.** M. Belgacem, M. Khatir, **M. A. Djehaf**, R. Bouddou, S. A. Zidi. Modeling and Control of Multi-Terminal Direct Current with Voltage Margin Control Strategy. *ICPEA.* DOI: 10.1109/ICPEA1.2019.8911168. [Link]

13. **2019.** Modular Multi-level Converter Based HVDC Systems Integrating Offshore Wind Farm. *International Symposium on Technology and Sustainable Industry Development.* [Link]

14. **2015.** A. Hameurlaine, S.-A. Zidi, A. Kouzou, **M. A. Djehaf**. Improvement of the HVDC link performances based on the protection functions. *IEEE ICIT.* DOI: 10.1109/ICIT.2015.7125498. [Link]

15. **2015.** Y. I. Djilani Kobibi, S. Hadjeri, **M. A. Djehaf**. Independent Power Flow Control and Dynamic Performance Enhancement by the UPFC. *ICEIT.* DOI: 10.1109/EITech.2015.7162970. [Link]

16. **2015. M. A. Djehaf**, S.-A. Zidi, Y. I. Djilani Kobibi, S. Hadjeri. Modeling of a multi-level converter based VSC HVDC supplying a dead load. *ICEIT.* DOI: 10.1109/EITech.2015.7162967. [Link]

17. **2013.** Y. I. Djilani Kobibi, S. Hadjeri, S. A. Zidi, **M. A. Djehaf**. Modelling a unified power flow controller for the study of power system steady state and dynamic characteristics. *ICMSAO.* DOI: 10.1109/ICMSAO.2013.6552627. [Link]

Other Conference Papers

1. **2019.** *Performance Analysis of a MTDC Transmission System with Master-Slave Control under Steady State and Faulted Conditions.* CIREI, Oran.
2. **2019.** *Dynamic Performance of Multi-Terminal VSC HVDC Systems with Master Slave Voltage Control.* ICMEMIS, Djelfa.
3. **2019.** *Voltage Stability Enhancement by the UPFC Using CPF Analysis.* Conference on smart electrical grids, Oran.
4. **2019.** *Contribution à la commande robuste d'une machine asynchrone à double alimentation appliquée au système éolien* [Contribution to the robust control of a doubly-fed induction machine applied to a wind energy system; in French]. Conference on smart electrical grids, Oran.
5. **2019.** *Modelling and Control of Multi-Terminal Direct Current VSC-HVDC Using DC Voltage Droop Control.* Doctoral days in electrical engineering.
6. **2019.** *FOC of Dual Star Induction Generator Used in Wind Energy Conversion System.* NSGE, El Oued.
7. **2019.** *L'apport des dispositifs FACTS dans l'amélioration du flux de puissance et du profil de tension* [The contribution of FACTS devices to improving power flow and voltage profile; in French]. Mascara.
8. **2018.** *Performance Analysis and Limitation of Master/Slave Voltage Control Strategy for Multi-Terminal HVDC System.* Doctoral days in electrical engineering.
9. **2016.** Y. I. Djilani Kobibi, S. Hadjeri, **M. A. Djehaf**. *Power Flow and Modal Analysis of a Power System Including Unified Power Flow Controller.* ICEET, Paris. DOI: 10.5281/zenodo.1338836. [Link]
10. **2015.** Y. I. Djilani Kobibi, **M. A. Djehaf**. *Incorporating the UPFC in Large-Scale Power System for the Study of Power Flow and Voltage Profile.* IMCAS, Salerno. DOI: 10.13140/RG.2.1.1425.9682. [Link]
11. **2015.** *Comparative Study between Active and Reactive Power Control of a Doubly-Fed Induction Generator in Wind Turbines.* ICEIT, Marrakech.
12. **2014.** *Modelling a UPFC for the Study of Power System Steady State and Dynamic Characteristics.* ICPEED, Oran.
13. **2013.** *Improvement of HVDC Control System to Mitigate Commutation Failure after AC System Asymmetric Faults.* CIGE, Béchar.
14. **2013.** *Dynamic Performance of VSC HVDC Link Supplying a Passive Load.* ICEES, Annaba.
15. **2013.** *Steady-State and Dynamic Performance of Asynchronous Back-to-Back VSC HVDC Link.* EPECS, Istanbul. [Link]
16. **2013.** *A Fuzzy Logic Controller for Reactive Power and PQ Bus Voltage Control.* EPECS, Istanbul. [Link]
17. **2013.** *Steady-State and Dynamic Modeling of Asynchronous Back-to-Back HVDC Link with Voltage Source Converters.* ICMSAO, Hammamet. [Link]
18. **2012.** *Dynamic Performance of Asynchronous Back-to-Back VSC HVDC Link.* ICEE, Skikda.
19. **2012.** *Modeling a UPFC for the Study of Power System Steady State and Transient Characteristics.* ICEE, Skikda.

20. **2012.** *Incorporating a Unified Power Flow Controller Model in Power Flow Calculation and Study of Dynamic Characteristics.* CMIGLE, Oran.
21. **2012.** *Dynamic Performance of Asynchronous Back-to-Back HVDC Link with Voltage Source Converters.* CMIGLE, Oran.
22. **2012.** *The Analysis of Commutation Failure and Protection Strategies in an HVDC System.* ICEE, Algiers.
23. **2012.** *Protection Strategy to Mitigate Commutation Failures in an HVDC System.* ICIPEE, Tébessa.

International Collaborations

- **IRC-ESTP, Cachan campus, France.** Collaboration and scientific exchanges with the ESTP research team, established at the invitation of Mohammed-Hichem Benzaama in 2026. Joint work covers physics-informed neural networks (PINNs) and their application to smart grids, state-of-charge estimation for battery racks, and the use of PINNs to predict heat and mass transfer in bio-based walls. This collaboration led to the publication *A deep residual graph convolutional neural network for hybrid physics data-driven modelling of latent heat transfer in cement mortar walls filled with bio-based phase change materials* (*Applied Thermal Engineering*, 2026, DOI: 10.1016/j.applthermaleng.2026.132512), co-authored with M.-H. Benzaama, A. Boudenne and K. Benzarti.
- **ESTACA, France (ESTACA'Lab, S2ET team: Embedded Systems and Energy for Transportation).** Collaboration on the thermal management of power-electronics modules for electric vehicles, at the interface of electrothermal modelling, reliability and electric mobility. It has produced two conference papers in 2026, *A Comparative Analysis of Cooling Techniques for IGBT power modules in electric vehicles* (CoDIT 2026) and *Étude comparative des techniques de refroidissement pour les modules de puissance IGBT pour véhicules électriques* (JCGE 2026), with Cherif Larouci, Mohammed Walid Omari, Nassim Rizoug and Cristina Morel. It also includes the co-supervision of the doctoral thesis of Mohammed Walid Omari (see the Supervision section).
- **University of Liège, Belgium.** Exchanges with the Smart Grids team of Prof. Damien Ernst at the Montefiore Institute. Part of the doctoral thesis was carried out in Liège; the collaboration concerns smart electrical grids and energy systems.

Research Interests and Funded Projects

- Member of the research laboratory ICEPS (*Intelligent Control and Electrical Power Systems*), team 02 "HVDC and Optimisation".
- Modelling and control of modular multilevel converters; HVDC-VSC and HVDC-CSC grids; multi-terminal direct-current grids and DC voltage control strategies.
- Integration of wind and photovoltaic power; transient and voltage stability; FACTS and UPFC; optimisation and intelligent control of energy systems.
- CNEPRU project *HVDC-VSC links for the connection of wind farms*, code J020212014002; project leader: Prof. Samir Hadjeri.
- PRFU project *Optimisation of the energy flows of photovoltaic plants through direct-current links: the case of the Algerian Sahara*, code A01L07UN220120190001.
- PRFU project *Integration of renewable energies into the Algerian power grid through multi-terminal DC (MTDC) networks*, code A01L07UN220120230002.

Education and Mobility

- **March 2012–January 2017: PhD in Electrical Engineering**, option power systems, Djillali Liabès University. Thesis: *Contribution to the analysis of an HVDC-VSC link based on multilevel converters*. Supervisor: Prof. Sid Ahmed Zidi; co-supervisor: Prof. Damien Ernst. Grade: very good (*très bien*); laboratory ICEPS.
- **July 2009–September 2011: Master's degree in Electrical Engineering**, option operation and management of power systems, Djillali Liabès University. Dissertation defended on 21 June 2011: *Study and simulation of a high-voltage direct-current link*. Supervisor: Prof. Sid Ahmed Zidi; laboratory ICEPS. Grade: very good (*très bien*).
- **September 2006–July 2009: Bachelor's degree in Electrical Engineering**, option electrical energy transmission and distribution, Djillali Liabès University. Dissertation defended in June 2009: *Optimisation of an electrical company: application to K.C.A.* Supervisor: Prof. Mostafa Brahami. Grade: very good (*très bien*).
- **2006: Scientific baccalaureate**, Okbi Ali high school, Sidi Bel Abbès; natural and life sciences stream; grade: fairly good (*assez bien*).
- **2014 and 2015: mobility internships**, Montefiore Institute, University of Liège, Belgium, in the team of Prof. Damien Ernst.
- **June 2015: Exceptional National Programme (PNE) scholarship**, Ministry of Higher Education and Scientific Research, Algeria.
- **September 2015–June 2016: long-term Erasmus stay and visiting researcher**, Montefiore Institute, University of Liège, Belgium; part of the thesis work was carried out there.
- **May 2023: mobility internship**, Technical University of Munich (TUM), Germany.

Additional Professional Experience and Internships

- **September 2016–January 2017:** account manager, ICT-Towers.
- **2010–2011:** work experience at EURL LABAIR Services, a private electrical contractor under agreement with Sonelgaz.
- **2009:** work experience at Établissements Brahami, an electrical contractor under agreement with Sonelgaz.
- **April–May 2009:** final-year internship at K.C.A.; **December 2008:** practical internship at the same company.
- **March 2009:** internship at the Zerwala dispatching substation (220/65/30 kV), Algerian electricity transmission company.

Teaching

Courses taught 2023–2026

Year	Module and audience	Format
2023–2024	Feedback Control Systems, third-year bachelor's (L3) in electromechanics	First semester; lectures, tutorials and laboratory sessions.
	Industrial Process Control, L3 electromechanics	Second semester; lectures, tutorials and laboratory sessions.
	Electrical Safety, L3 electromechanics	Lectures.
	Electrical Safety, second-year bachelor's (L2) in automatic control	Lectures.
	Industrial Electrical Networks, first-year master's (M1) in electromechanics	Lectures.
2024–2025	Feedback Control Systems, third-year bachelor's (L3) in electromechanics	First semester; lectures, tutorials and laboratory sessions.
	Industrial Process Control, L3 electromechanics	Second semester; lectures, tutorials and laboratory sessions.
	Electrical Safety, L3 electromechanics	Lectures.
	Electrical Safety, second-year bachelor's (L2) in automatic control	Lectures.
	Industrial Electrical Networks, first-year master's (M1) in electromechanics	Lectures.
2025–2026	Feedback Control Systems, third-year bachelor's (L3) in electromechanics	First semester; lectures, tutorials and laboratory sessions.
	Industrial Process Control, L3 electromechanics	Second semester; lectures, tutorials and laboratory sessions.
	Electrical Safety, L3 electromechanics	Lectures.
	Electrical Safety, second-year bachelor's (L2) in automatic control	Lectures.
	Industrial Electrical Networks, first-year master's (M1) in electromechanics	Lectures.

Lectures and doctoral training courses

Module	Years and audience
Feedback Control Systems	2018–2019 to 2021–2022; third-year bachelor's in electromechanics.
Industrial Process Control	2017–2018 to 2021–2022; third-year bachelor's in electromechanics.
Applied Numerical Methods	2019–2020; first-year electromechanics; also taught online.
Control of Electromechanical Drives	2017–2018; third-year electromechanics.
Applications of Artificial Intelligence in Electrical Engineering: Neural Networks and Fuzzy Logic	2021–2022; first-year PhD in electrical engineering.

Tutorials and laboratory sessions

Module	Format and years
Feedback Control Systems	Tutorials and laboratory sessions, 2018–2019 to 2021–2022; tutorials also in electrical engineering in 2019–2020.
Industrial Process Control	Tutorials and laboratory sessions, 2017–2018 to 2021–2022.
Advanced Power Electronics	Laboratory sessions, 2018–2019 to 2021–2022.
Intelligent Control	Laboratory sessions, 2019–2020 to 2021–2022; master's in automatic control.
Artificial Intelligence Techniques	Laboratory sessions, 2018–2019 to 2020–2021; electrical engineering.
Applied Numerical Methods	Laboratory sessions, 2019–2020.
Linear Systems Control	Laboratory sessions, 2018–2019; automatic control.
Final-Year Project	Laboratory sessions and project supervision, 2018–2019 to 2020–2021.
Power System Analysis Tools	Lectures, tutorials and laboratory sessions; part-time lecturer, 2012–2017.
Numerical Methods	Lectures and laboratory sessions; part-time lecturer.
Power System Protection	Laboratory sessions; part-time lecturer.
Electrical Engineering	Laboratory sessions; part-time lecturer.
Power System Modelling	Laboratory sessions; part-time lecturer.
Computer Science	Laboratory sessions; part-time lecturer.
Signal Theory	Tutorials; part-time lecturer.

Teaching Materials

1. **Feedback Control Systems: course notes**, 103 pages, 2022, for third-year bachelor's students in electromechanics. Posted on the university e-learning platform on 9 October 2022: course link.
2. **Intelligent Control: laboratory manual**, 62 pages, 2022, for second-year master's students in automatic control. Posted on 9 October 2022: course link.

Supervision and Research Training

Master's theses

Year	Topic	Students
2021–2022	Intelligent control of underactuated nonlinear systems	Bousahra Benabdellah; Benbahou Abdelouahheb
2021–2022	Reinforcement learning for the control of nonlinear systems	Kafi Ahmed Amine; Chikh Miloud Mohamed
2020–2021	Stabilisation and intelligent control of an inverted pendulum	Feddaoui Anis; Boukhari Yasser Alaa Eddine
2020–2021	Reinforcement learning applied to the control of electromechanical actuators	Fouires Sara

2020–2021	Contribution of artificial intelligence to the energy management of a microgrid	Amadou Adamou Abdourahim
2019–2020	Study and implementation of an advanced controller for a DC motor	Elghouteb Med El Hattab; Boubout Ali
2019–2020	Advanced control of a modular multilevel converter applied to reactive power compensation	Benchinoune Mouâd; Kemmoun Abdelhak
2018–2019	Contribution to the modelling and control of a modular multilevel converter applied to reactive power compensation	Benoua Imene

Co-supervised doctoral theses

- **Since 2025: co-supervision** of Mohammed Walid Omari, thesis *Electrothermal modelling and thermal management of the power electronics of an electric-vehicle traction chain*. Enrolled in the PhD programme on 3 February 2025, Université Paris-Saclay, doctoral school *Electrical, Optical, Bio-physics and Engineering*; discipline: electrical engineering; laboratory ESTACA'Lab, S2ET team (Embedded Systems and Energy for Transportation); host faculty: Faculté des sciences d'Orsay. Thesis director: Cherif Larouci.
- **2017–2022: co-supervision** of Belgacem Moussa, thesis *Contribution to the study of meshed direct-current grids for future smart electrical grids*; thesis director: Prof. Mohamed Khatir.

Bachelor's projects

- **2020–2021:** modelling and control of various static converters, third-year electromechanics project.
- **2019–2020:** design of an inverted pendulum on a cart, third-year electromechanics project.
- **2018–2019:** design of a level-control system, third-year electromechanics project.
- **Before 2017:** study and control of MPPT in wind energy conversion systems, bachelor's project.
- **Before 2017:** study of a hybrid battery–supercapacitor storage system, bachelor's project.
- **Before 2017:** study and modelling of power transmission cables, bachelor's project.
- **Before 2017:** modelling of static converters for electric power transmission, bachelor's project.

Scientific Service, Skills and Languages

Peer review of manuscripts for scientific journals, including *IFAC Journal of Systems and Control* and *International Journal of Electrical Power and Energy Systems*; member of scientific committees of international conferences.

Further training: scientific writing with LaTeX (Oran partner of the Campus numérique francophone, May 2017); intensive English B1+/B2 (ISLV, University of Liège, February–April 2016); real-time simulation with RT-LAB/OPAL-RT (January 2013); two certifications in ICT for education and pedagogy (University of Constantine).

Tools: MATLAB/Simulink, PSCAD/EMTDC, EMTP-RV, RT-LAB/OPAL-RT (ARTEMiS, RT-Events), COMSOL Multiphysics, Maple, C/C++, Pascal, Microsoft Office, Windows and Linux environments.

Languages: Arabic (native), French (fluent), English (scientific working proficiency).

Community engagement: during the Covid-19 pandemic, participation in an association assisting sick, disabled, elderly or unemployed people.